DRAFT STANDARD OPERATING PROCEDURE

Sugarcane Germplasm Characterization

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-GC-SOP-02	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To produce reproducible botanical, morphological, and selected agronomic descriptors for sugarcane germplasm so that accessions can be identified, compared, catalogued, screened for duplicates, and rationally selected as breeding parents.

2. Scope

Applies to established germplasm accessions and reference cultivars maintained in field collections or characterization nurseries. Characterization is distinct from multi-environment performance evaluation: highly heritable descriptors establish identity, while yield, quality, disease, and stress-response traits belong to linked evaluation datasets.

3. Responsibilities

- Germplasm curator: defines descriptor list and reference standards, approves final accession profile.
- Breeder/taxonomist: scores morphological descriptors and resolves abnormal/off-type plants.
- Field technician: maintains uniform crop management, labeling, sampling, and photographs.
- Laboratory staff: performs Brix/juice-quality or molecular tests when included.

4. Field standardization

- Use healthy, true-to-type planting material of common physiological age.
- Use a common planting date and agronomic management wherever practical.
- Include reference/check cultivars that express known descriptor states.
- Avoid scoring severe border-affected, diseased, mechanically damaged, or atypically late tillers unless the trait itself is the subject of scoring.
- Repeat uncertain descriptors in a second crop cycle, ratoon, or season before making a permanent database decision.

5. Sampling strategy

For quantitative descriptors, sample 5-10 representative mature stalks per accession where sufficient material is available. Record the exact sampling position and crop age used by the institution so repeated measurements are comparable. USDA sugarcane registration work commonly reports botanical means across multiple representative stalks; a 10-stalk sample is a practical reference for mature clonal material.

6. Characterization procedure

1. Verify accession identity and field map before entering the plot. Photograph the accession label in the first image.
2. Score whole-plant characters at a defined crop age, preferably near maturity before harvest: canopy habit, stool compactness, erectness/lodging, tillering, flowering, and trashing behavior.
3. Select representative mature stalks and measure stalk height, stalk diameter, internode length, and related quantitative characters using the same anatomical reference position for all entries.
4. Clean only the amount of wax/trash needed to observe rind color and node characters. Record exposed and unexposed rind color separately because sunlight can change pigmentation.
5. Describe internode shape, waxiness, growth cracks, root band, growth ring, root primordia, bud shape, bud groove/furrow, bud position, and germ pore.
6. Describe leaf blade, sheath, ligule, auricle, dewlap, pubescence, and related leaf characters using standardized terminology and reference images.
7. Take standardized photographs of whole stool/canopy, mature stalk, node and bud, exposed/unexposed internode, leaf sheath, and dewlap/ligule/auricle. Include a scale in close-ups.
8. Record evaluation traits such as Brix, Pol, sucrose, fibre, yield, disease resistance, or stress tolerance in a linked evaluation sheet rather than overwriting core characterization descriptors.
9. Flag plants that conflict with the expected accession profile. Do not immediately rogue an off-type until identity verification is completed.
10. Have a second trained scorer verify rare or ambiguous descriptor states before final database entry.

7. Recommended descriptor groups

Descriptor group	Examples	Operational note
Whole plant	Growth habit; canopy architecture; stool compactness; erectness; tillering; lodging; flowering	Score at standardized crop age
Stalk	Height; diameter; internode length/shape; exposed/unexposed color; wax; growth cracks; pithiness	Use defined internode position
Node / root band	Growth ring width; root band width; root primordia; node prominence	Use close-up photo with scale
Bud	Shape; size; color; position; groove/furrow; cushion; germ pore; pubescence; projection above growth ring	Score several nodes
Leaf blade	Length; width; attitude; color; midrib	Avoid senescent leaves
Leaf sheath	Color; wax; pubescence; adherence/trashing	Use mature green leaves
Ligule / auricle / dewlap	Shape; size; color; pubescence	Standard botanical terminology
Evaluation - separate dataset	Brix; Pol; sucrose; fibre; millable stalks; yield; ratooning; disease/stress reaction	Environment-dependent; record trial metadata

8. Photographic standard

- Use a neutral or consistent background for close-ups where practical.
- Place accession ID and ruler/scale in the frame; do not rely on image filename alone.
- Use the same camera orientation and approximate distance for comparable images.
- Photograph both sun-exposed and covered rind when color is a descriptor.
- Archive original files with accession ID, date, crop, field, and photographer metadata.

9. Molecular identity support

Where facilities are available, establish a DNA reference sample for priority accessions and use validated SSR/SNP/genomic markers to resolve synonyms, duplicates, and suspected mixtures. Molecular fingerprints complement, but do not replace, field morphology and source documentation.

10. Quality-control and acceptance criteria

- At least 90% of mandatory descriptor fields completed for well-established material; unresolved states flagged, not guessed.
- Quantitative measurements include sample size and crop age.
- Characterization and evaluation variables stored in separate but linked data tables.
- Off-types are tagged and investigated before removal.
- Final profile reviewed by a second trained scorer or curator for high-value accessions.

11. Characterization field sheet

Section	Field / observation
Identity	Accession no.; variety name; field/row/plot; date; crop age; scorer
Whole plant	Growth habit; stool compactness; lodging; flowering; tillering

Section	Field / observation
Stalk	Height; diameter; internode length; shape; wax; rind colors; cracks
Node/Bud	Growth ring; root band; bud shape; bud furrow; germ pore; pubescence
Leaf	Blade length/width; attitude; sheath pubescence; ligule; auricle; dewlap
Photos	Whole stool; stalk; bud/node; sheath; ligule/auricle/dewlap
Evaluation link	Brix / sucrose / fibre / yield / disease dataset ID
Verification	Second scorer / curator review; notes

References and technical basis

USDA-ARS sugarcane registration example using GRIN botanical descriptors - Web source

ICAR-Sugarcane Breeding Institute - maintenance and characterization of world collection - Web source

Sugar Regulatory Administration - research portfolio including morpho-agronomic characterization - Web source